

Aptitudinal Alignment Institutional Fidelity Field Theory

Aptitude Systems

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Introduction

Aptitudinal Alignment measures how closely a public institution’s documented conduct aligns with what that institution is capable of and obligated to achieve. It is designed as a reverse-FICO system for public entities: instead of scoring an individual’s credit risk, every eligible *entity* — a judge, court, agency, office, or institution — receives one explainable *Alignment Score* derived from the public record.

Each public record, decision, or procedural act is treated as a discrete unit of institutional behavior. These units interact through measurable fields of fidelity, consistency, transparency, efficiency, equity, and accountability. The emergent field defines the Alignment Score.

Product Language

- The page and system name is **Aptitudinal Alignment**.
- The single entity result is the **Alignment Score**.
- The six explainable components are **dimension scores**.
- The interpreted category is the **Alignment Status**.
- Each scored subject is an **entity** with a versioned **potential profile**.

The term “Composite Score” is not a product term. Wherever source material said “Aptitude Score” or “composite,” the unified manual says “Alignment Score.”

Design Principles

- The quantum model is an analytical metaphor and data abstraction — not literal quantum computation.
- Public institutional records are the primary and sufficient evidentiary basis.
- A score is never invented when evidence thresholds are not met; the result is `insufficient_data` with explicit reasons.
- Raw inputs and derived outputs are preserved separately so every score can be recomputed.
- Legal-intelligence conclusions and institutional scores are separate bounded products.
- No Alignment Score is a finding of individual wrongdoing, a prediction of future conduct, or a substitute for an official finding.

Canonical Vocabulary

One vocabulary is used across the codebase, the API, and this manual.

Term	Meaning
Evidence item	A sourced institutional event, record, decision, filing, or report.
Source record	Citation and provenance metadata for an evidence item: source type, credibility tier, date, jurisdiction, and retrieval details.
Behavioral quantum	The normalized analytical representation of an evidence item. It is a data abstraction, not a physical particle.

Term	Meaning
Metric observation	A raw measurement mapped to a dimension and sub-metric.
Dimension score	A reproducible score for Fidelity, Consistency, Transparency, Efficiency, Equity, or Accountability.
Alignment Score	The universal FICO-like result for an entity: a weighted, explainable measure of how closely documented conduct approaches that entity's defined potential after evidence and hard-floor checks.
Entity	Any subject evaluated by the system: judge, court, agency, office, or other institution.
Potential profile	The versioned standard for what an entity can and should achieve, based on mandate, role, jurisdiction, peer class, resources, and public obligations.
Alignment Status	The interpreted category of the Alignment Score: <code>high_alignment</code> , <code>adequate</code> , <code>at_risk</code> , <code>low_alignment</code> , <code>critical_failure</code> , <code>insufficient_data</code> , or <code>suppressed</code> .

Dimension Symbols

The six dimensions use unambiguous symbols in code and documentation: *F* Fidelity, *C* Consistency, *T* Transparency, *E* Efficiency, *Q* Equity, and *Ac* Accountability. *Q* means Equity; *Ac* means Accountability. No other meaning is attached to these symbols.

Evidence and Provenance Model

Both the legal-intelligence pipeline and the Alignment scoring pipeline use one shared evidence foundation. Each evidence item identifies the institution or actor, the event, the source record, source type and credibility tier, citation, document date, jurisdiction, dimension and sub-metric mapping, raw and normalized values, uncertainty, and the processing, ontology, metric registry, and normalization versions that produced it.

Raw evidence is immutable. Derived observations and scores are append-only revisions linked to the input hashes and configuration versions used to produce them.

Source Type Allowlist

Only records with these `source_type` values are eligible for institutional score computation:

```
public_record, regulatory_filing, court_decision, agency_report,
cfpb_complaint, ftc_action, inspector_general, individual_submission
```

Records with `source_type = individual_submission` are supplementary (credibility tier 6, weight 0.05) and cannot dominate any dimension unless corroborated by a public record. Their absence never degrades institutional score validity.

Provenance Fields

```
{
  "id": "hash(entityId + eventId + sourceRecordId)",
  "entity_id": "uuid",
  "event_id": "uuid",
```

```

"source_record_id": "uuid",
"source_type": "court_decision",
"credibility_tier": 1,
"citation": "State v. Example, 123 Or App 45 (2024)",
"document_date": "2024-03-01",
"jurisdiction_code": "OR",
"dimension": "F",
"raw_value": 0.91,
"normalized_value": 0.91,
"confidence": 1.0,
"processing_version": "brain-v1",
"ontology_version": "ontology-v1",
"metric_registry_version": "metrics-v1",
"normalization_version": "norm-v1",
"input_hash": "hex",
"recorded_at": "ISO-8601"
}

```

Legal Intelligence Pipeline

The legal-intelligence pipeline transforms lived legal experiences into structured, queryable, auditable legal intelligence. It is a separate bounded product from institutional scoring.

Flow

```

narrative -> structured event -> claims/doctrines
          -> authority graph and retrieval -> procedural pathways

```

Agents

- **Intake Parser:** structures observable facts only; never infers legal conclusions.
- **Legal Classifier:** maps to claims and doctrines using the ontology only; conservative classification.
- **Graph Seeder:** builds the legal knowledge graph with typed edges (supports, governed_by, codified_in, violates, resolved_by).
- **Vector Retrieval:** returns ranked supporting authority with metadata — supporting material only, no conclusions.
- **Procedural Engine:** builds ordered enforcement pathways with prerequisites and jurisdiction alignment.

Invariants

- No output without provenance.
- No citation without source.
- No authority without jurisdiction.
- No overwrite of historical records.
- No legal conclusion without supporting authorities.
- An incident never silently becomes an institutional score.

Aptitudinal Alignment Scoring

Flow

```
entity + potential profile + evidence
-> metric observations
-> normalization
-> six dimension evaluators
-> hard floors
-> Alignment Score
-> tier / status
```

Every supported entity receives the same explainable score contract, but not the same metric registry. A judge, court, agency, prosecutor's office, or credit reporting agency gets a potential profile appropriate to its mandate, role, jurisdiction, peer class, resources, and public obligations.

Universal Entity Contract

```
{
  "entity_id": "uuid",
  "entity_type": "judge | court | agency | office | institution",
  "potential_profile_version": "string",
  "alignment_score": 0.0,
  "alignment_status": "high_alignment | adequate | at_risk | low_alignment
    | critical_failure | insufficient_data | suppressed",
  "dimensions": { "F": 0.0, "C": 0.0, "T": 0.0, "E": 0.0, "Q": 0.0, "Ac":
    0.0 },
  "evidence_summary": { "count": 0, "source_types": [], "
    missing_requirements": [] },
  "flags": [],
  "audit_reference": "string"
}
```

The numeric range is normalized to [0,1] internally. A FICO-like integer scale may be used only through a documented presentation transform; it must never hide the six dimensions, evidence sufficiency, or limitations.

Default Weights

$$A = \sum_{i=1}^6 w_i \cdot D_i \quad \sum_{i=1}^6 w_i = 1.0$$

Dimension	Symbol	Default weight	Rationale
Fidelity	<i>F</i>	0.22	Legal mandate adherence is foundational
Consistency	<i>C</i>	0.18	Variance in outcomes signals systemic bias
Transparency	<i>T</i>	0.17	Public record access is a legal right
Efficiency	<i>E</i>	0.15	Resource stewardship proxies competence
Equity	<i>Q</i>	0.16	Demographic fairness is a constitutional floor
Accountability	<i>Ac</i>	0.12	Self-correction capacity determines trajectory

Classification Tiers

Tier	Score range	Alignment Status
I	0.85 – 1.00	high_alignment
II	0.70 – 0.84	adequate
III	0.50 – 0.69	at_risk
IV	0.30 – 0.49	low_alignment
V	0.00 – 0.29	critical_failure

Evidence Thresholds and Insufficient Data

A score is emitted only when the configured evidence thresholds pass: at least 12 behavioral quanta per dimension from at least 3 distinct source types. When any dimension fails its threshold, the result is `insufficient_data`: `alignment_score` is null, the failing dimensions are null, and `evidence_summary.missing_requirements` explains why. The system never estimates a missing dimension.

Hard Floors

Equity hard floors override the weighted average and cannot be bypassed:

- Finding of intentional discrimination within 10 years $\Rightarrow Q_{\min} = 0.15$
- Active consent decree for a civil rights violation $\Rightarrow Q_{\min} = 0.25$
- Pattern-or-practice finding $\Rightarrow Q_{\min} = 0.20$

All floor applications are recorded in the score record before the Alignment Score is computed.

Normalization

$$D_{\text{norm}} = \frac{D_{\text{raw}} - D_{\min}}{D_{\max} - D_{\min}}$$

Bounds are configurable per entity class. Values are clamped to $[0, 1]$ — no extrapolation permitted.

Reproducibility

A reproducibility record stores input hashes, configuration versions, code version, and output hash. Any score can be re-derived byte-for-byte from the quantum set and the potential profile version.

Six Dimensions and Metric Registry

Each dimension is defined once. The operational sub-metric tables, weights, source lists, and agent specifications come from the Beacon Scoring Engine specification (source: `Bias Beacon Documentation.tex`, the second of the two LaTeX manuals on the maintainer’s desktop), reproduced verbatim where they define behavior. Only the rules that change scoring behavior are restated here; full metric tables live in the source specification and its metric registry.

Fidelity (F)

Measures whether an institution adheres to its legal mandate, enabling statutes, and constitutional obligations. Sub-metrics: Legal Mandate Compliance Rate (LMCR), Policy Adherence Score (PAS), Regulatory Violation Reversal Rate (RVR), and Statutory Override Frequency (SOF, inverted). Consent orders count as negative compliance evidence regardless of whether the institution agreed to them. Self-reported compliance documentation receives a 0.50 credibility discount unless corroborated. A single court finding of *ultra vires* action floors LMCR at 0.20 for the affected period.

Consistency (C)

Measures variance in institutional outcomes across substantively similar cases. Sub-metrics: Outcome Variance Index (OVI, inverted), Precedent Adherence Rate (PAR), Cross-Case Disparity Score (CDS, inverted), and Decision Reversal Rate (DRR, inverted). A demographic disparity above 15% in any protected class triggers an equity cross-flag regardless of the overall C score. Institutions under consent decrees carry permanent low-consistency markers until a 5-year clean record is established.

Transparency (T)

Measures whether the institution makes its records, decisions, and procedures accessible. Sub-metrics: Record Disclosure Rate (RDR), Response Time Index (RTI), FOIA/Public Records Compliance Rate (FCR), and Proactive Access Index (PAX). Response time is measured from institutional *receipt*, not acknowledgment. Institutions requiring in-person filing only receive an automatic PAX deduction of 0.10. An institution held in contempt for disclosure failure within the last 5 years cannot score above $T = 0.40$.

Efficiency (E)

Measures how effectively an institution converts public resources into mandated outcomes. Sub-metrics: Case Resolution Rate (CRR), Processing Time Score (PTS), Resource Utilization Index (RUI, net of litigation costs), and Backlog Clearance Rate (BCR). Administrative closures without documented outcome do not count. Efficiency scores are always displayed with the peer-class benchmark visible. Budget cuts exceeding 20% in a single year trigger a documented 0.10 grace adjustment for that year only.

Equity (Q)

Measures whether the institution delivers its mandate fairly across demographic groups, geographic areas, and economic strata. Sub-metrics: Demographic Disparity Index (DDI, inverted), Geographic Access Score (GAS), Protected Class Outcome Rate (PCOR), and Appeals Success Rate for protected classes (ASR). Hard floors are defined in Section 5. A minimum $n = 100$ per demographic cell is required before statistical claims; the system never imputes demographic data. A Tier V equity score triggers a mandatory regulatory referral workflow regardless of the overall Alignment Score.

Accountability (Ac)

Measures the institution's capacity to identify its own failures, correct them, and institutionalize reforms. Sub-metrics: Self-Correction Rate (SCR, with a $1.5\times$ multiplier for proactive correction), Audit Compliance (AC), Complaint Resolution Rate (CRR_c, distinct from Efficiency CRR), and Remediation Action Rate (RAR). A pattern of recurring identical violations (3+ consecutive periods) caps Ac at 0.35. Complaints marked resolved without documented resolution trigger a RESOLUTION_FRAUD flag and an audit of all complaint-resolution scores for the preceding 24 months.

Source Credibility and Normalization Bounds

Credibility Hierarchy

Conflicting quanta from multiple sources are resolved by tier weight:

Tier	Weight	Source description
1	1.00	Court orders, consent decrees, regulatory enforcement actions
2	0.90	IG reports, GAO findings, examination summaries, sworn testimony
3	0.75	Aggregated complaint databases, formal complaints
4	0.50	Self-reported public filings (cross-validation required)
5	0.20	Third-party research and reporting (corroboration required)
6	0.05	Individual submissions — supplementary only

Normalization Bounds

Normalization bounds are configured per entity class in the metric registry (e.g., the NCRA peer class bounds table from `Bias Beacon Documentation.tex` for credit reporting agencies). Every bound records its calibration rationale. Values are clamped to $[0, 1]$ — no extrapolation permitted.

Governance and Reproducibility

- Public institutional records are the primary and sufficient evidentiary basis; individual submissions are supplementary.
- Every claim and score component carries a citation chain.
- Raw evidence is immutable; derived outputs are append-only revisions.
- Hard floors are applied and logged before the Alignment Score is computed.
- A reproducibility record stores input hashes, configuration versions, code version, and output hash.
- No Alignment Score is a finding of individual wrongdoing, a prediction of future conduct, or a substitute for an official finding.

Implementation Phases

The following phases are tracked in the repository as they are built. The current implementation status is recorded at the top of each phase.

1. Shared evidence foundation and provenance contracts (built).
2. Universal Alignment Score engine with insufficient-data and hard floors (built).
3. Judiciary filter experience with case-type filtering and the scale-and-rotate panel (built).
4. Identity system: gold A mark, hero first letter, favicon (built).
5. CRA and institutional data connectors (specified).
6. Normalization service with per-class bounds (specified).

7. Public score API (specified).
8. Reproducibility audit tooling (specified).